

DiscoGMPI Numerical Implementation Notes

DiscoGMPI project

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Contents

1 Polynomial Approximation Space

Each scalar component of $\mathbf{E}$ and $\mathbf{H}$ is approximated independently on every tetrahedron by a polynomial in

$$\mathbb{P}^N(K) = \{p(x, y, z) : \deg p \leq N\}. \quad (1)$$

The number of volume nodes is

$$N_p = \frac{(N+1)(N+2)(N+3)}{6}, \quad (2)$$

and the number of nodes on a triangular face is

$$N_{fp} = \frac{(N+1)(N+2)}{2}. \quad (3)$$

The Maxwell state stores six arrays of size $N_p \times K$, where K is the number of rank-local tetrahedra including ghosts in distributed runs.

2 Reference Tetrahedron and Nodal Sets

The reference tetrahedron has vertices

$$(-1, -1, -1), \quad (1, -1, -1), \quad (-1, 1, -1), \quad (-1, -1, 1). \quad (4)$$

`build_reference_tet(N)` constructs equispaced nodal points on this reference element. The current nodal set is deterministic but not an optimized warp-blend or Fekete set, so interpolation conditioning should be watched at high order.

An orthonormal modal tetrahedral basis is evaluated at the nodes to form the Vandermonde matrix

$$V_{ij} = \phi_j(r_i, s_i, t_i). \quad (5)$$

The inverse Vandermonde maps nodal values to modal coefficients and represents nodal basis functions as modal combinations.

3 Quadrature and Cubature

Independent diagnostic integration uses Jaskowiec–Sukumar tetrahedral cubature rules through `get_JaskowiecSukumar_cubature(order)`. Production diagnostics commonly use

$$q = \max(2, 2N + 4), \quad (6)$$

so field errors, energy-density errors, chirality, and momentum diagnostics are not dominated by low-order quadrature.

4 Mass Matrices

On the reference tetrahedron, the nodal mass matrix is

$$M = V^{-T}V^{-1}, \quad (7)$$

because the modal basis is orthonormal. For an affine physical element,

$$M_K = |J_K|M. \quad (8)$$

The DG energy diagnostic uses this mass matrix directly:

$$\mathcal{H}_h = \frac{1}{2} \sum_K |J_K| \left[\varepsilon \sum_a E_{a,K}^T M E_{a,K} + \mu \sum_a H_{a,K}^T M H_{a,K} \right]. \quad (9)$$

5 Derivative and Stiffness Matrices

Reference derivative matrices are formed from modal derivative Vandermonde matrices:

$$D_r = V_r V^{-1}, \quad D_s = V_s V^{-1}, \quad D_t = V_t V^{-1}. \quad (10)$$

The weak stiffness matrices are

$$S_r = M D_r, \quad S_s = M D_s, \quad S_t = M D_t. \quad (11)$$

Metric terms transform these into physical derivative matrices D_x, D_y, D_z and weak operators S_x, S_y, S_z . The Hesthaven–Warburton path uses strong derivative operators for volume curls. The Poisson-bracket path uses weak operators for one partition and transposed weak operators for the complementary partition.

6 Face Operators and Connectivity

Reference face operators identify the N_{fp} nodes on each triangular face and build face mass/lifting operators. A face residual contributes conceptually as

$$M_K^{-1} M_{f,K} f \quad (12)$$

to the nodal right-hand side.

DG topology is constructed by sorting the three vertex IDs of each tetrahedral face. A face appearing once is a physical boundary face, a face appearing twice is an interior face, and a face appearing more than twice is rejected as non-manifold. Interior trace maps store minus and plus element IDs, local face IDs, face-node lists, plus-to-minus nodal permutations, outward normals, and face areas.

7 Distributed Ownership and Periodic Maps

In MPI runs, each rank owns a subset of global tetrahedra and stores a one-face halo. Volume work and diagnostics use owned elements only. Ghost fields are updated by exchanging face traces before surface flux evaluation.

Periodic boundaries are directed face pairs. The pairing logic matches boundary face centroids after a translation and constructs node permutations to align plus-side and minus-side nodes. For the default rectangular periodic box, the boundary tags are

$$\begin{aligned} 1 &\leftrightarrow 2 && \text{x-min} / \text{x-max} \\ 3 &\leftrightarrow 4 && \text{y-min} / \text{y-max} \\ 5 &\leftrightarrow 6 && \text{z-min} / \text{z-max}. \end{aligned}$$

Periodic tags should be handled by the periodic exchange object and should remain unregistered as physical boundary conditions in the ordinary registry.

8 Metric Terms and Mesh Quality

For each affine tetrahedron, the Jacobian of the map from reference coordinates to physical coordinates gives

$$\nabla_x u = J_K^{-T} \nabla_{r,s,t} u. \quad (13)$$

The determinant $|J_K|$ scales volume integrals, and face geometry supplies normals and areas.

The convergence CSV records mesh-quality metrics: volume min/max/total/ratio, edge min/max/ratio, element-size min/max/ratio based on maximum edge length, and tetrahedron mean-ratio quality. The mean-ratio quality is

$$q_K = \frac{12(3V_K)^{2/3}}{\sum_{e \in K} |e|^2}, \quad (14)$$

where the denominator sums the squared lengths of the six tetrahedron edges.

9 Flux Evaluation

Flux evaluation proceeds by gathering minus and plus traces, aligning plus-side nodes to the minus-side ordering, evaluating the face correction, and lifting it into the volume. The centered Hesthaven–Warburton correction is

$$F_E^c = \frac{1}{2} \mathbf{n} \times (\mathbf{H}^+ - \mathbf{H}^-), \quad F_H^c = -\frac{1}{2} \mathbf{n} \times (\mathbf{E}^+ - \mathbf{E}^-). \quad (15)$$

The upwind correction adds impedance-weighted tangential penalties:

$$-\frac{Y}{2} \mathbf{n} \times (\mathbf{n} \times (\mathbf{E}^+ - \mathbf{E}^-)), \quad (16)$$

$$-\frac{Z}{2} \mathbf{n} \times (\mathbf{n} \times (\mathbf{H}^+ - \mathbf{H}^-)). \quad (17)$$

The Poisson-bracket operator uses

$$F_E^{PB} = -\frac{1}{2\varepsilon} \mathbf{n} \times (\mathbf{H}^- - \mathbf{H}^+), \quad F_H^{PB} = -\frac{1}{2\mu} \mathbf{n} \times (\mathbf{E}^- + \mathbf{E}^+). \quad (18)$$

Only the central Poisson-bracket flux is supported.

10 Boundary Trace Construction

PEC and PMC boundary conditions are handled by reflected exterior traces:

$$\mathbf{E}_{PEC}^+ = -\mathbf{E}^- + 2(\mathbf{n} \cdot \mathbf{E}^-) \mathbf{n}, \quad \mathbf{H}_{PEC}^+ = \mathbf{H}^-, \quad (19)$$

$$\mathbf{E}_{PMC}^+ = \mathbf{E}^-, \quad \mathbf{H}_{PMC}^+ = -\mathbf{H}^- + 2(\mathbf{n} \cdot \mathbf{H}^-) \mathbf{n}. \quad (20)$$

Periodic boundaries use true traces from paired periodic faces rather than reflected traces.

11 Time Integration

DiscoGMPI contains conventional explicit Runge–Kutta methods and explicit partitioned symplectic Runge–Kutta methods. For a semidiscrete ODE $\dot{U} = R(U)$, explicit RK methods use

$$U^{(i)} = U^n + \Delta t \sum_{j < i} a_{ij} K_j, \quad K_i = R(U^{(i)}), \quad U^{n+1} = U^n + \Delta t \sum_i b_i K_i. \quad (21)$$

For Poisson-bracket Maxwell runs, the recommended method is ESPRK with magnetic-first partitioning. The convergence drivers usually set ESPRK order to $N+1$ for DG order N so time error does not dominate the spatial study.

12 CFL Condition

The basic timestep estimate is

$$\Delta t = \text{CFL} \frac{\min_K h_K}{(2N+1)c}, \quad (22)$$

where c is the material wave speed. The CFL element size used by the timestep estimate is

$$h_K = \frac{3V_K}{A_K}, \quad (23)$$

with V_K the tetrahedron volume and A_K its total surface area. The final timestep is adjusted so the run lands exactly on the requested final time.

13 Mesh-Size Definitions

The code and CSV files distinguish several mesh-size measures:

- **target_h**: requested Gmsh size, typically L_x/N_x ;
- **characteristic_h**: measured volume scale, typically $(|\Omega|/N_{el})^{1/3}$;
- **h_min, h_max**: quality sizes based on maximum tetrahedron edge length;
- **CFL h_K** : $3V_K/A_K$.

Production convergence rates use the measured characteristic mesh size so structured and independently generated unstructured families are treated consistently.

14 Structured and Unstructured Families

The cavity convergence driver supports structured unit-cube meshes and unstructured unit-cube Gmsh families. The periodic convergence driver supports structured $[0, 2] \times [0, 1] \times [0, 1]$ periodic box meshes and unstructured periodic Gmsh families. The CSV stores **mesh_family** so post-processing can separate structured and unstructured studies.

15 Expected Convergence Rates

For polynomial degree N , the current regression targets are aggregate electric-field rate $N+1$ and aggregate magnetic-field rate N . Component errors and rates are recorded for

$$E_x, E_y, E_z, H_x, H_y, H_z. \quad (24)$$

Component rates are suppressed when the coarse error is at roundoff level, which avoids meaningless rates for analytically zero components. The production pass/fail checks compare observed rates with expected rates rather than testing only monotone error decrease.

16 Output Diagnostics

Convergence CSV files include DG order, ESPRK order, mesh family, mesh level, element count, owned-element distribution, mesh sizes, mesh quality, aggregate errors and rates, per-component errors and rates, available invariant diagnostics, and runtime. Production transient drivers additionally write time-series ParaView output, quadrature diagnostics, run metadata, partition metadata, integration-point samples, and checkpoints.